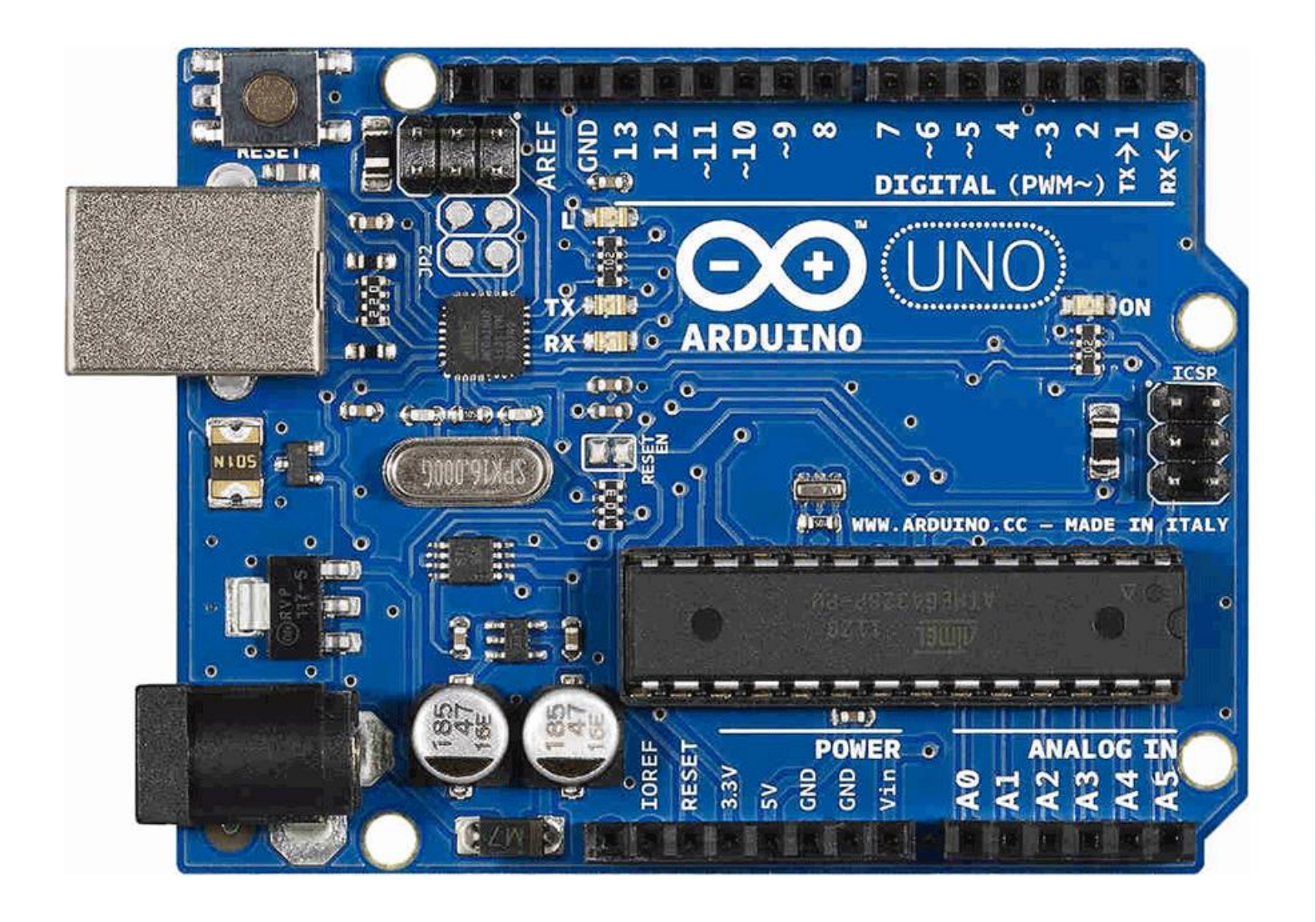
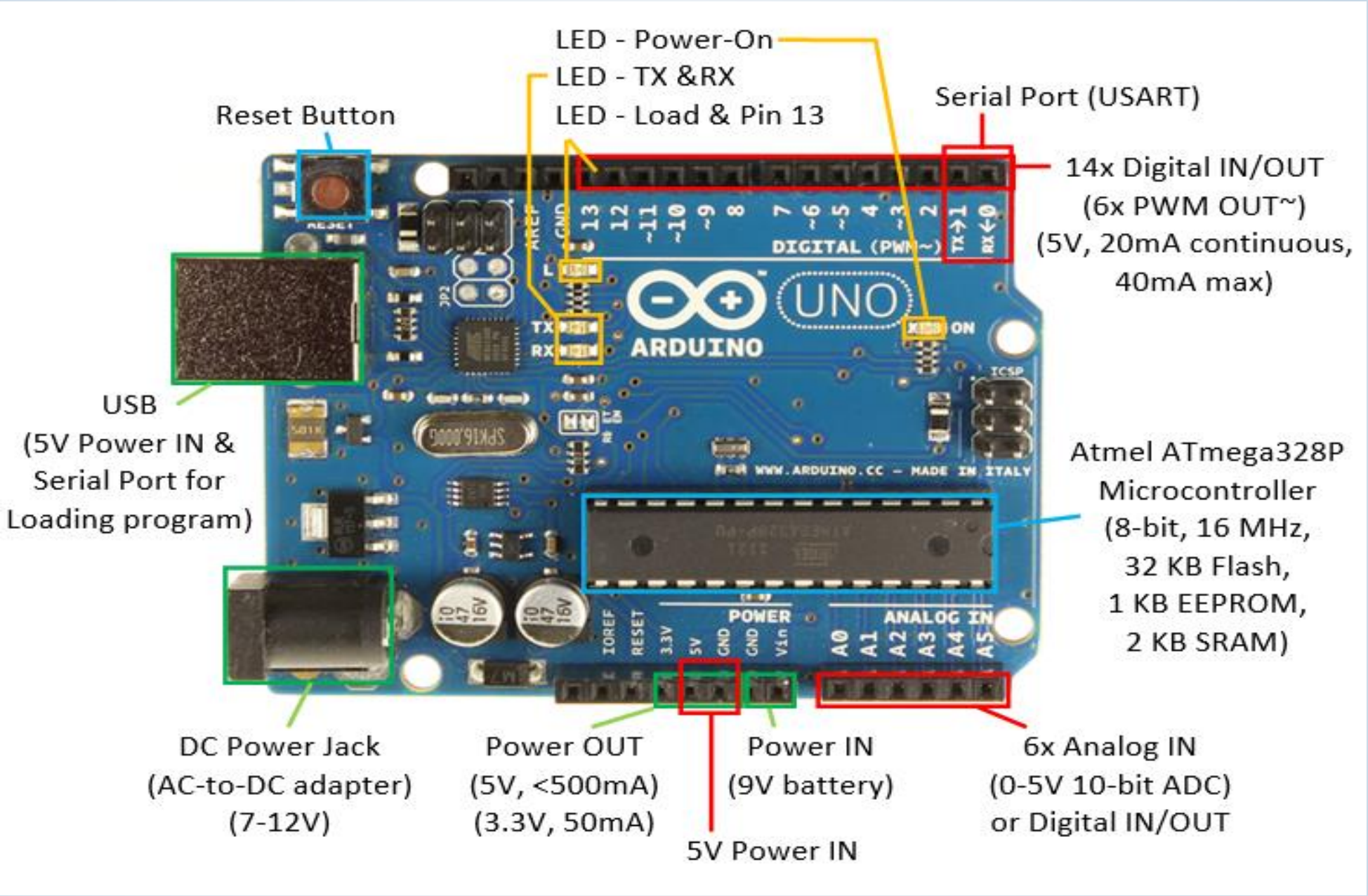
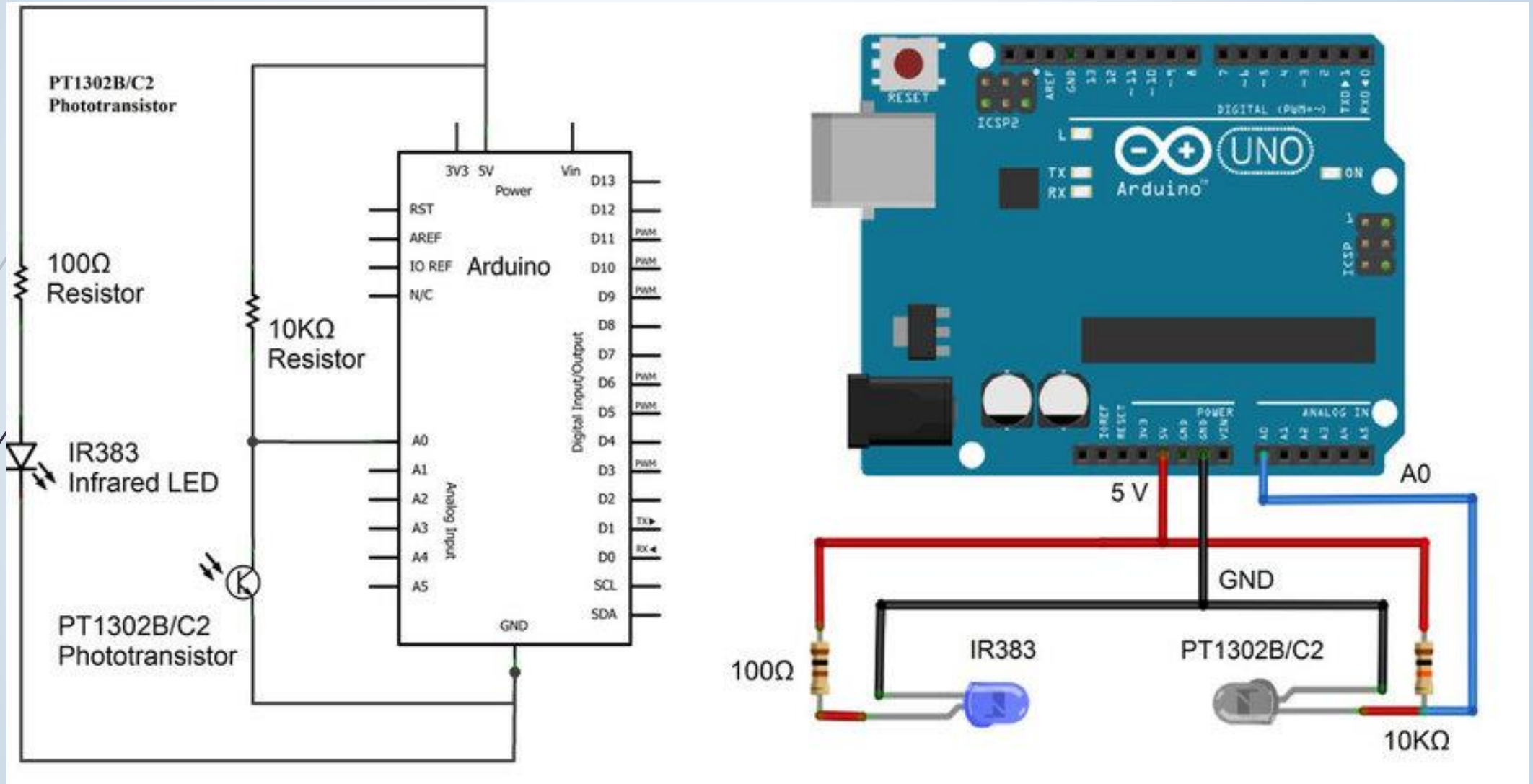
- Course from Edx: https://learning.edx.org/course/course-v1:CurtinX+IOT1x+2T2018/home
- Course from Edx: https://learning.edx.org/course/course-v1:CurtinX+IOT2x+2T2018/home
- Free Hardware and IoT Ebooks_ OReilly Media

- Documents at https://www.arduino.cc/

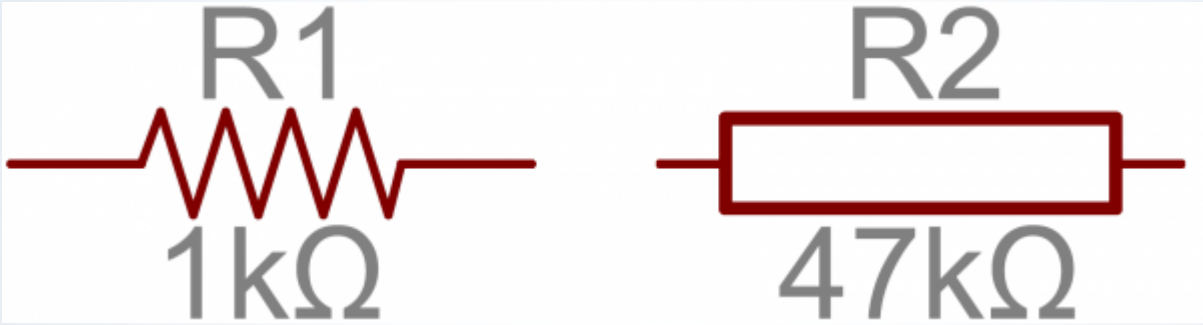
Arduino UNO


RESET
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
UNO
ARDUINO
ON
TX
RX
ICSP
RESET
WWW.ARDUINO.CC – MADE IN ITALY
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

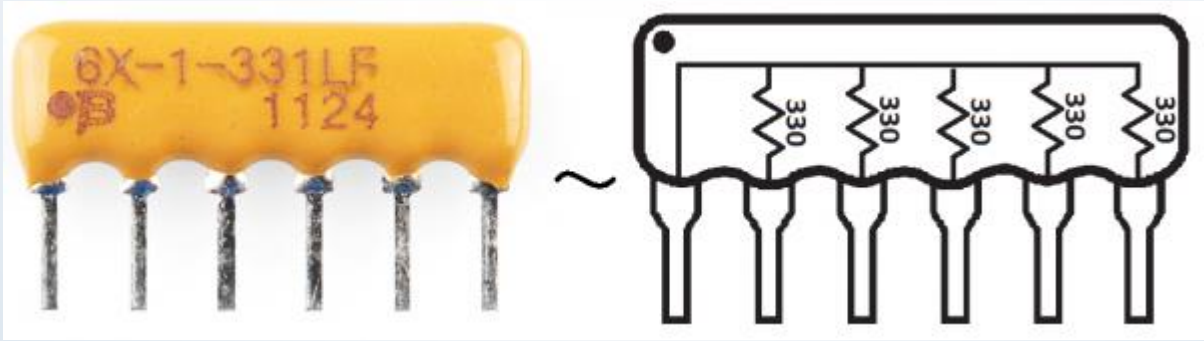

LED - Power-On
LED - TX &RX
LED - Load & Pin 13
Reset Button
Serial Port (USART)
14x Digital IN/OUT
(6x PWM OUT~)
(5V, 20mA continuous,
40mA max)
USB
(5V Power IN &
Serial Port for
Loading program)
Atmel ATmega328P
Microcontroller
(8-bit, 16 MHz,
32 KB Flash,
1 KB EEPROM,
2 KB SRAM)
DC Power Jack
(AC-to-DC adapter)
(7-12V)
Power OUT
(5V, <500mA)
(3.3V, 50mA)
Power IN
(9V battery)
6x Analog IN
(0-5V 10-bit ADC)
or Digital IN/OUT
5V Power IN
UNO
ARDUINO
DIGITAL (PWM~)
POWER
ANALOG IN
WWW.ARDUINO.CC – MADE IN ITALY

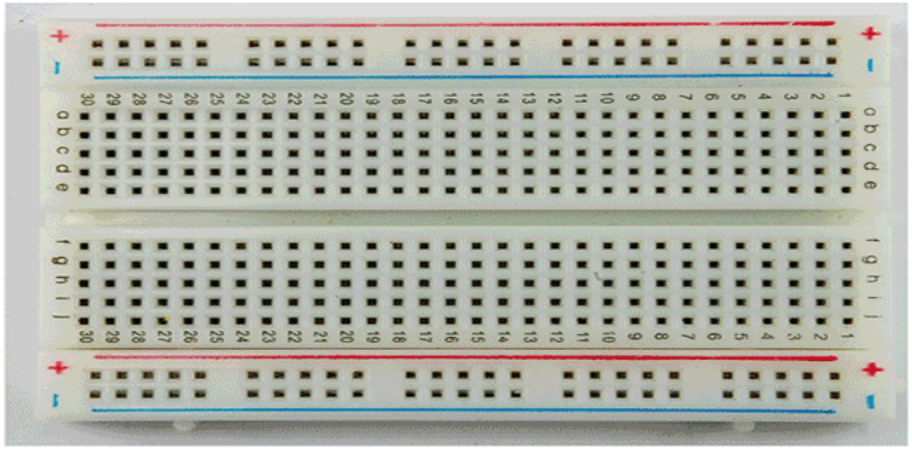

PT1302B/C2
Phototransistor
100Ω
Resistor
10KΩ
Resistor
IR383
Infrared LED
PT1302B/C2
Phototransistor
3V3 5V
Power
Vin
RST
AREF
IO REF
N/C
Arduino
A0
A1
A2
A3
A4
A5
Analog Input
GND
D13
D12
D11
D10
D9
D8
D7
D6
D5
D4
D3
D2
D1
D0
SCL
SDA
Digital Input/Output
PWM
TX
RX
RESET
ICSP2
L
TX
RX
UNO
Arduino
ON
ICSP
POWER
ANALOG IN
5 V
A0
GND
100Ω
IR383
PT1302B/C2
10KΩ

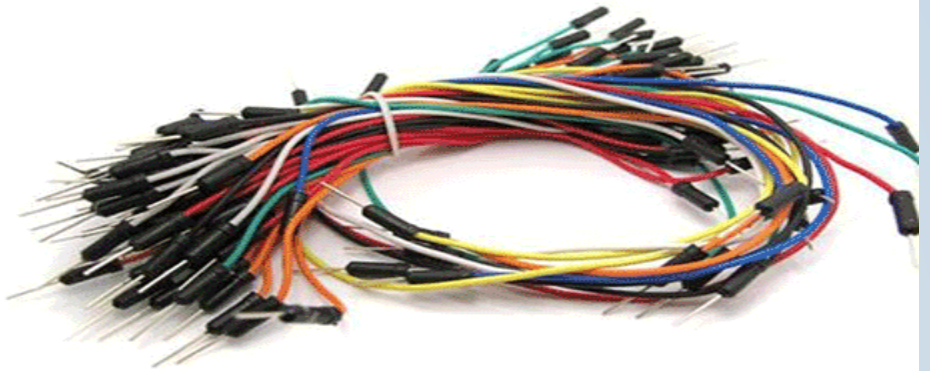
Resistors

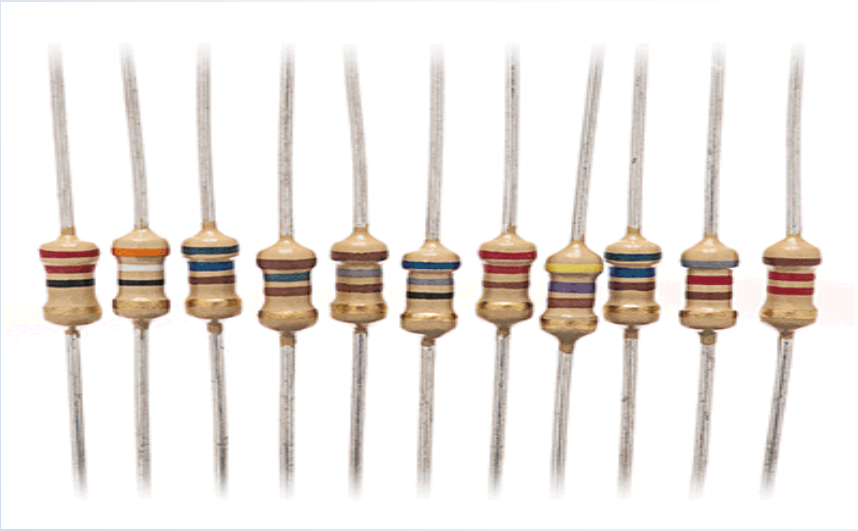


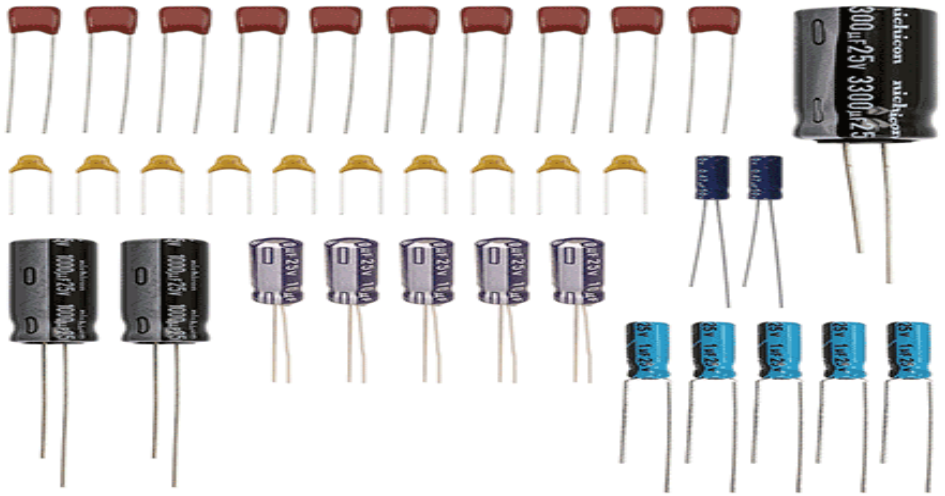






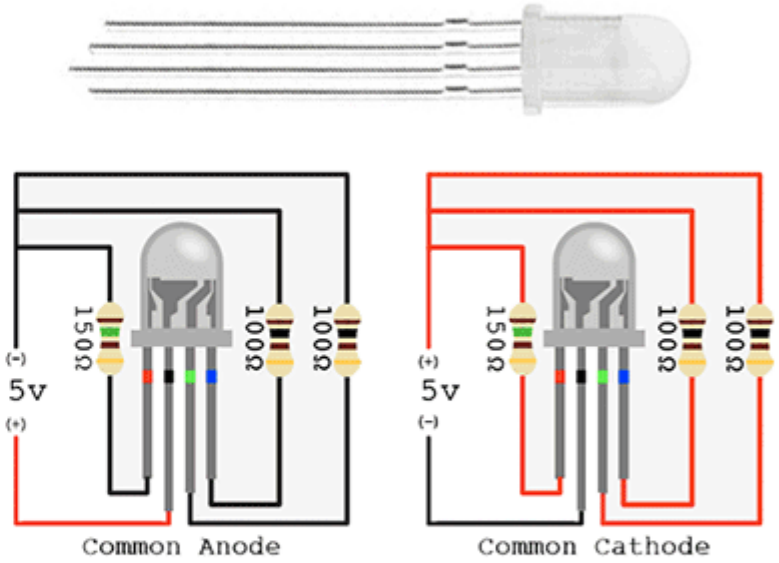





nichicon
3300μF25V

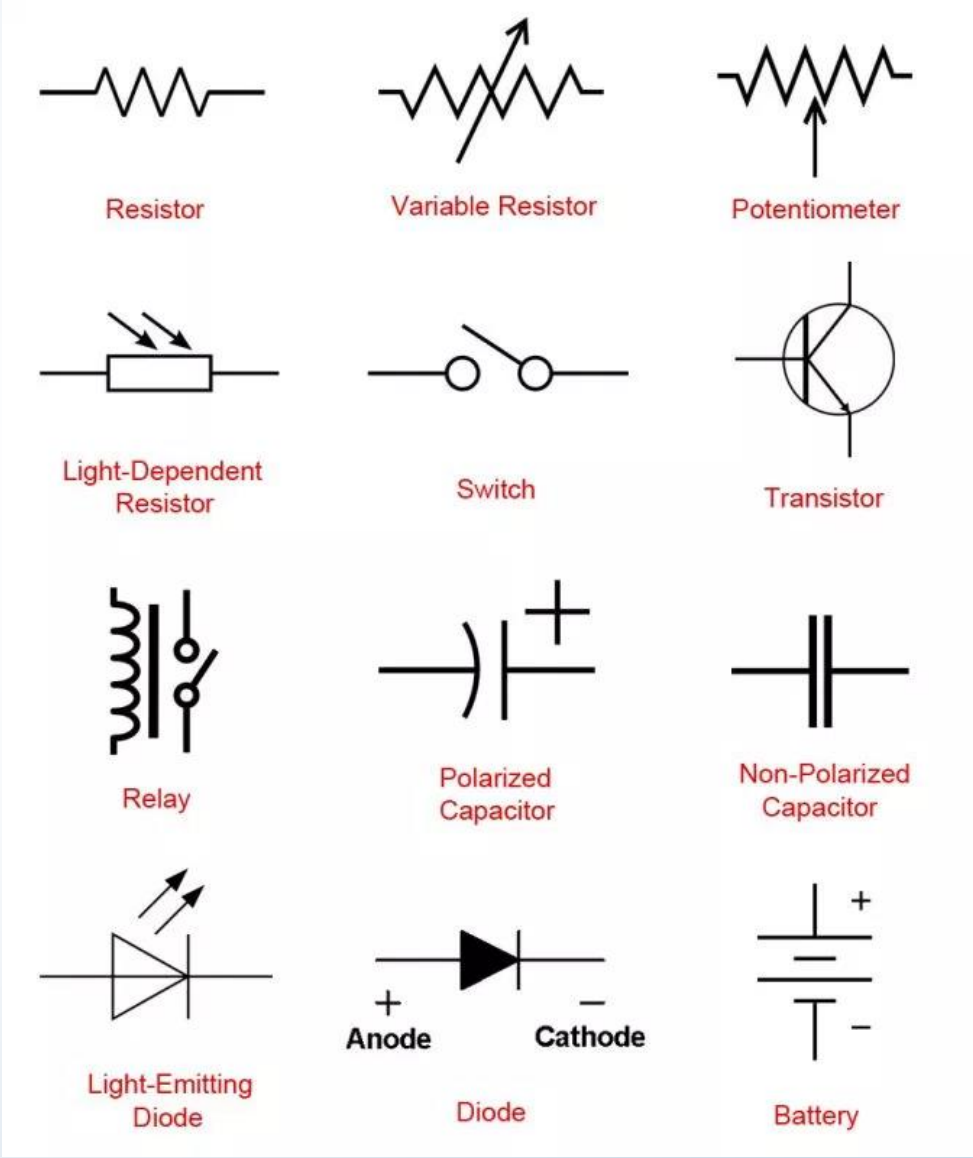


Simple LEDs



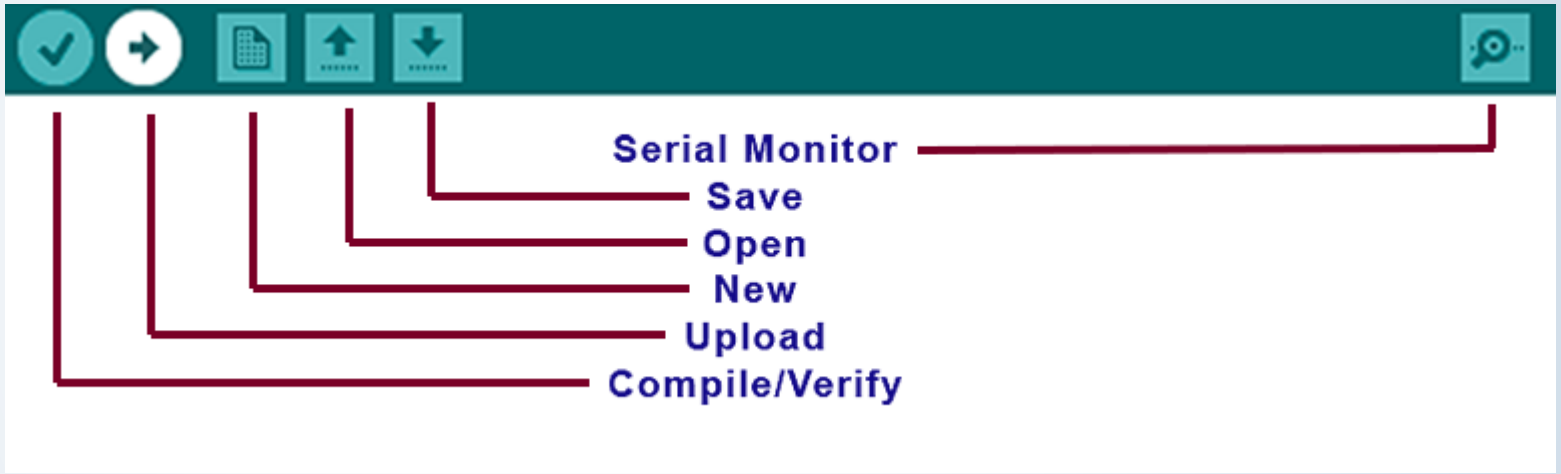
RGB LEDs

a few of the most commonly used electronic symbols in the US.




FLUKE 177 TRUE RMS MULTIMETER
0.000 DC V
AutoHOLD
HOLD
MINMAX
RANGE
mV
Ω
V
mA
A
OFF
400 mA
10A
FUSED
600V CAT IV
1000V CAT III
COM

PROGRAMMING ARDUINO - BASIC FUNCTIONS


Serial Monitor
Save
Open
New
Upload
Compile/Verify

ARDUINO LOGIC

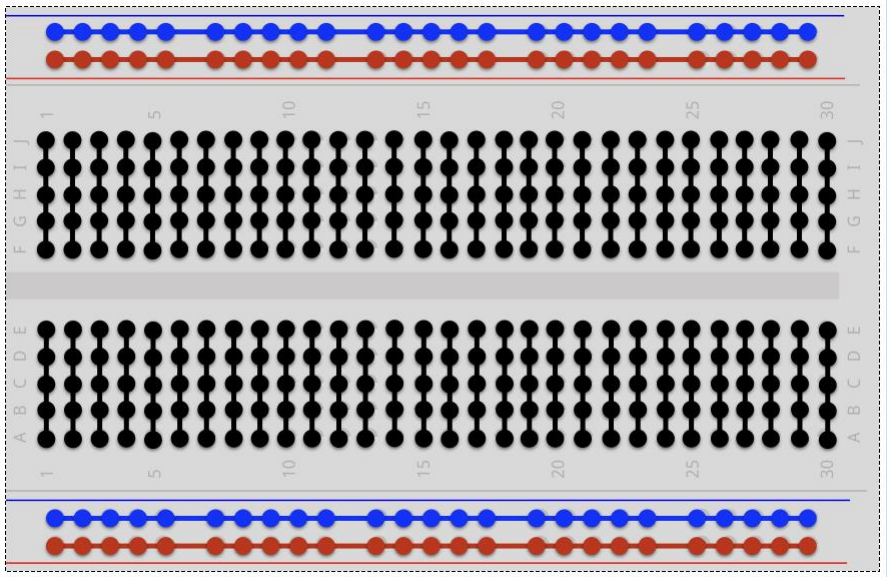
- • **setup ()** - here we put all the commands that must run once when the unit is turned on. Variable value initializations are usually entered and, in any case, the identification of the inputs/outputs that we will use
- • **loop ()** - here we write our program. The commands that exist will run and when it reaches the end, the loop () will resume, continuing from the start, and again. This will happen continuously, as long as Arduino is running or until the reset key is pressed.

LAB 1: blink LED

Hardware Required

1. Arduino UNO
2. Led 3mm
3. Resistor 330R
4. Breadboard
5. Wire

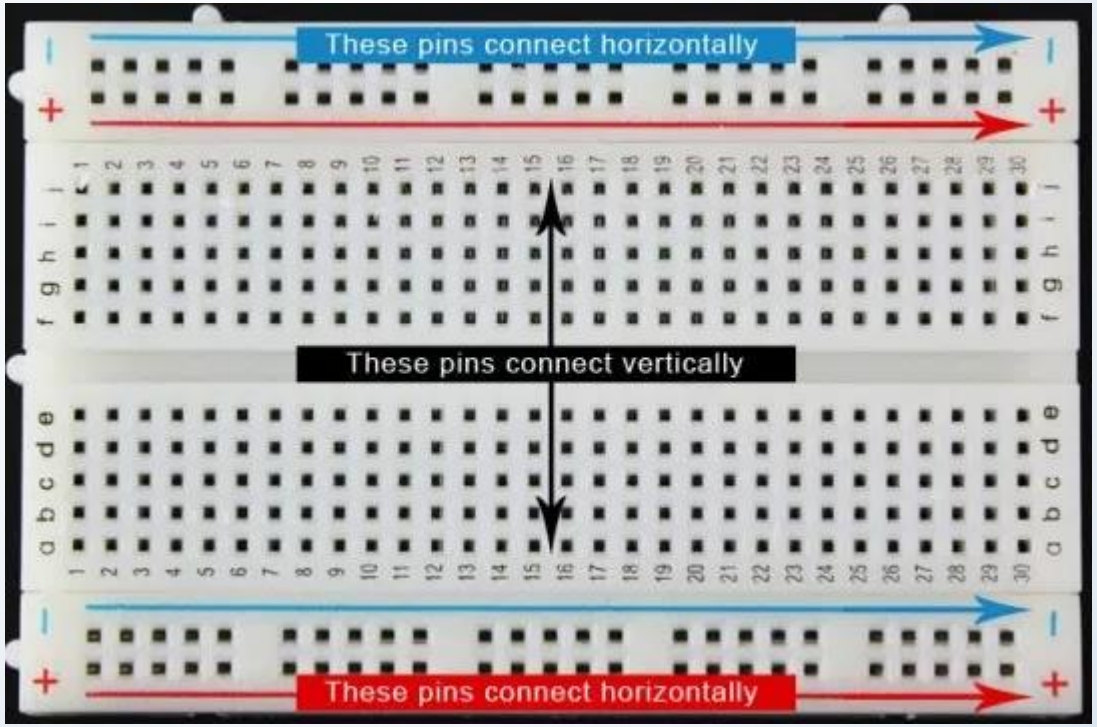


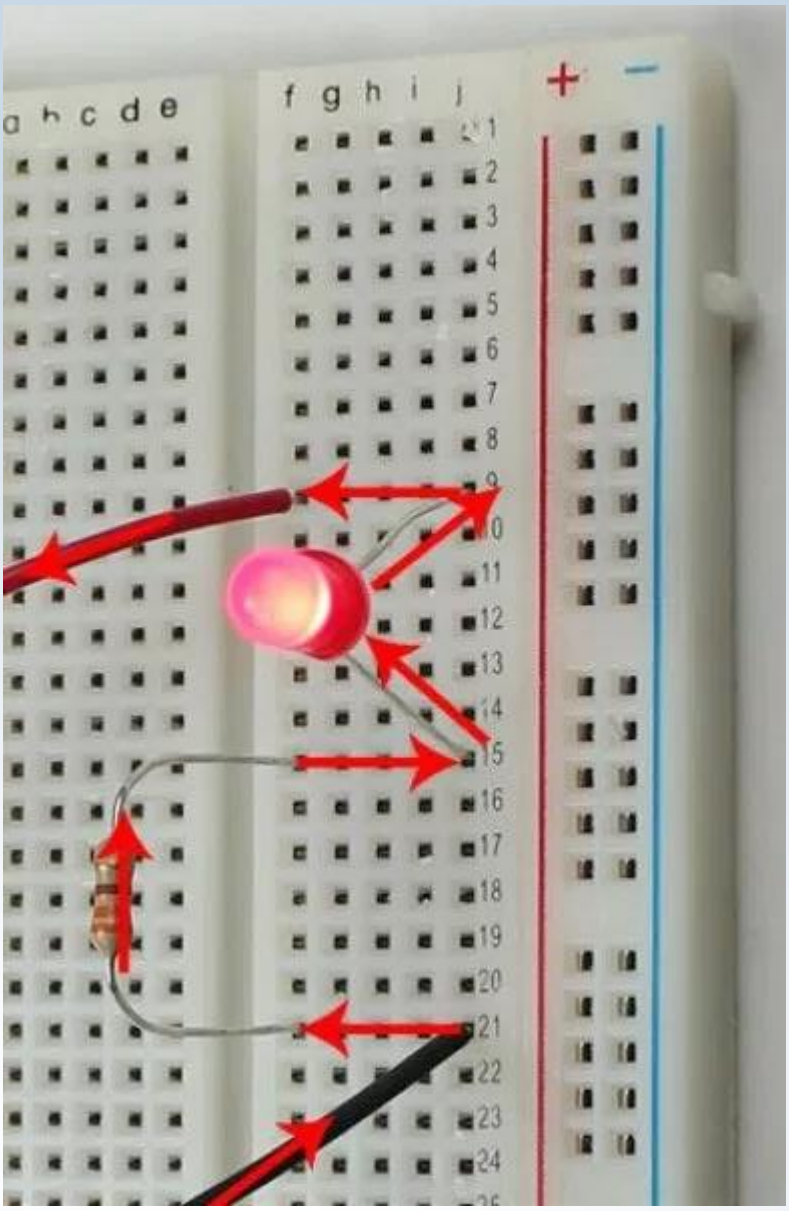


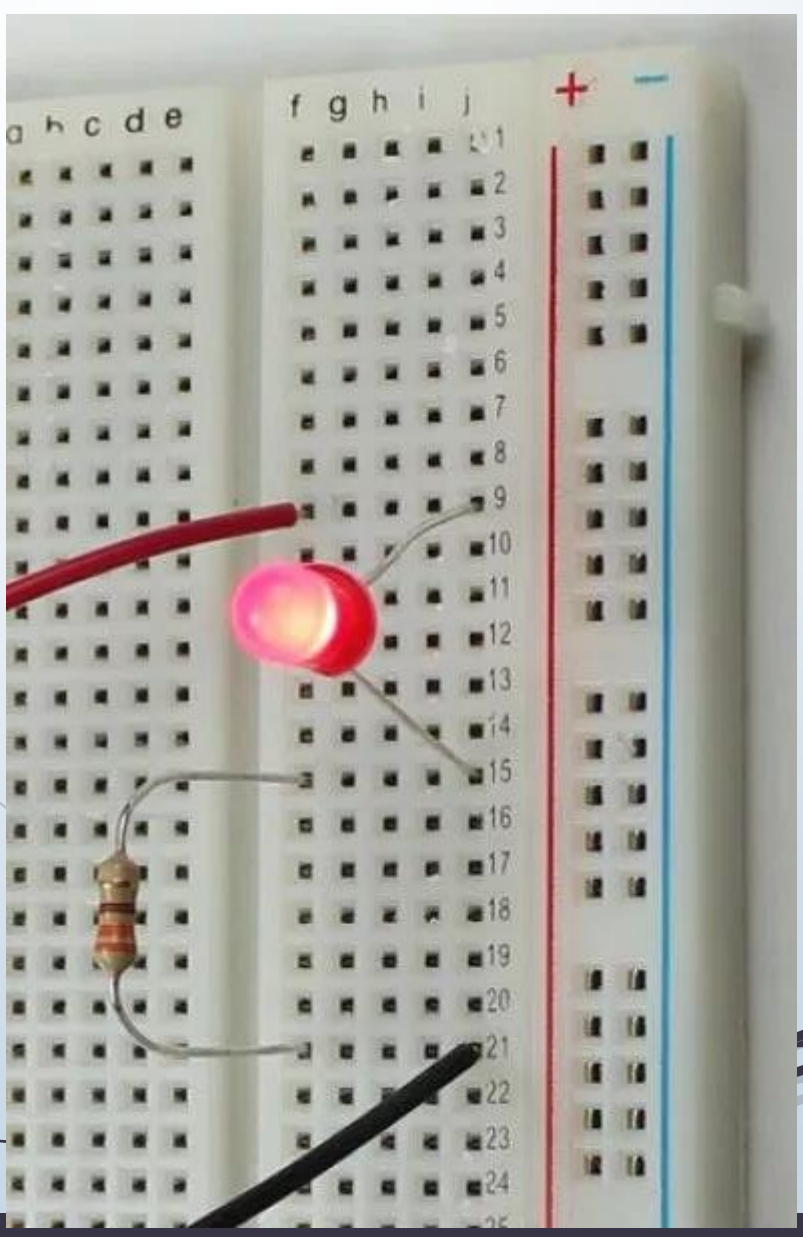


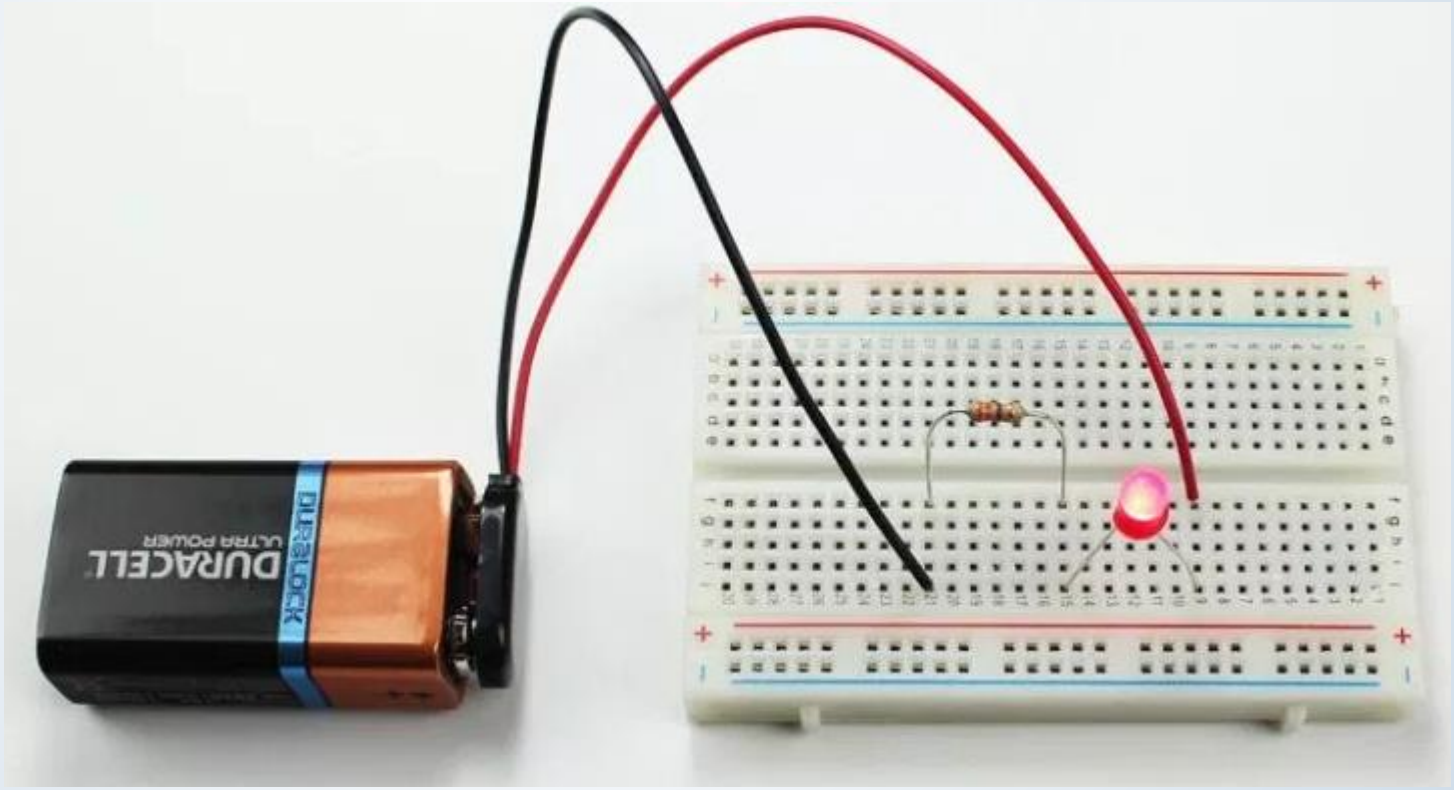





These pins connect horizontally
These pins connect vertically
These pins connect horizontally

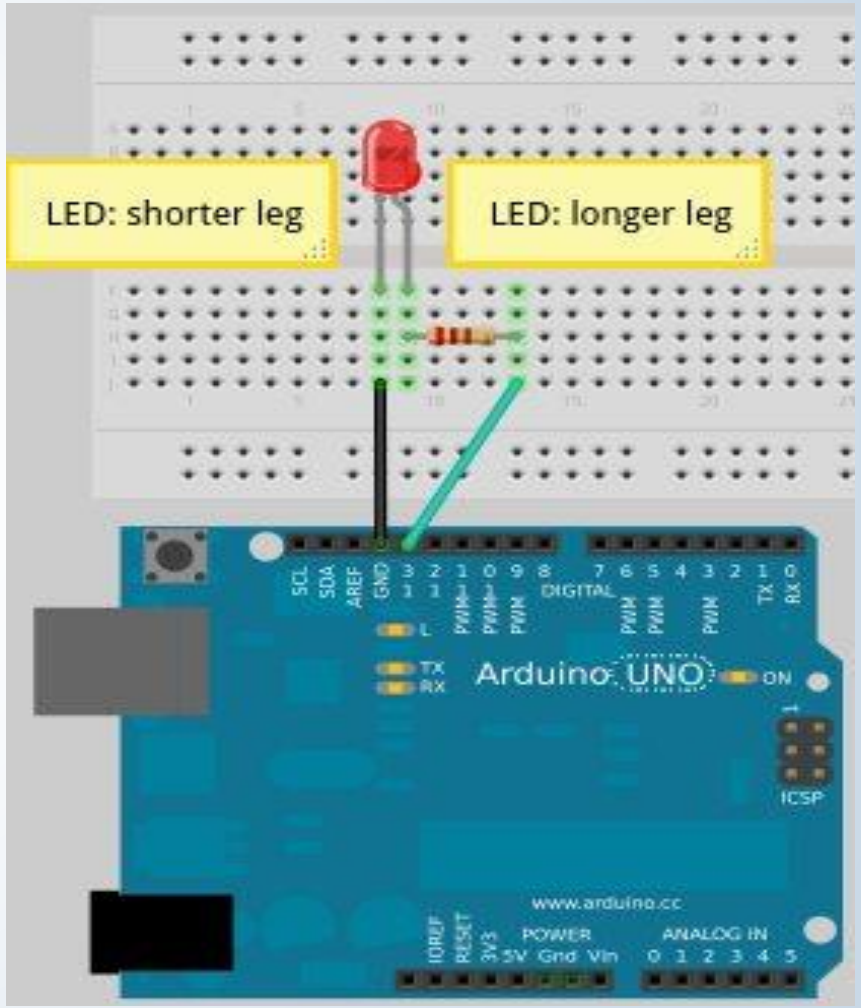

a b c d e
f g h i j
+
−


a b c d e
f g h i j
+
−


DURACELL
ULTRA POWER
DURALOCK

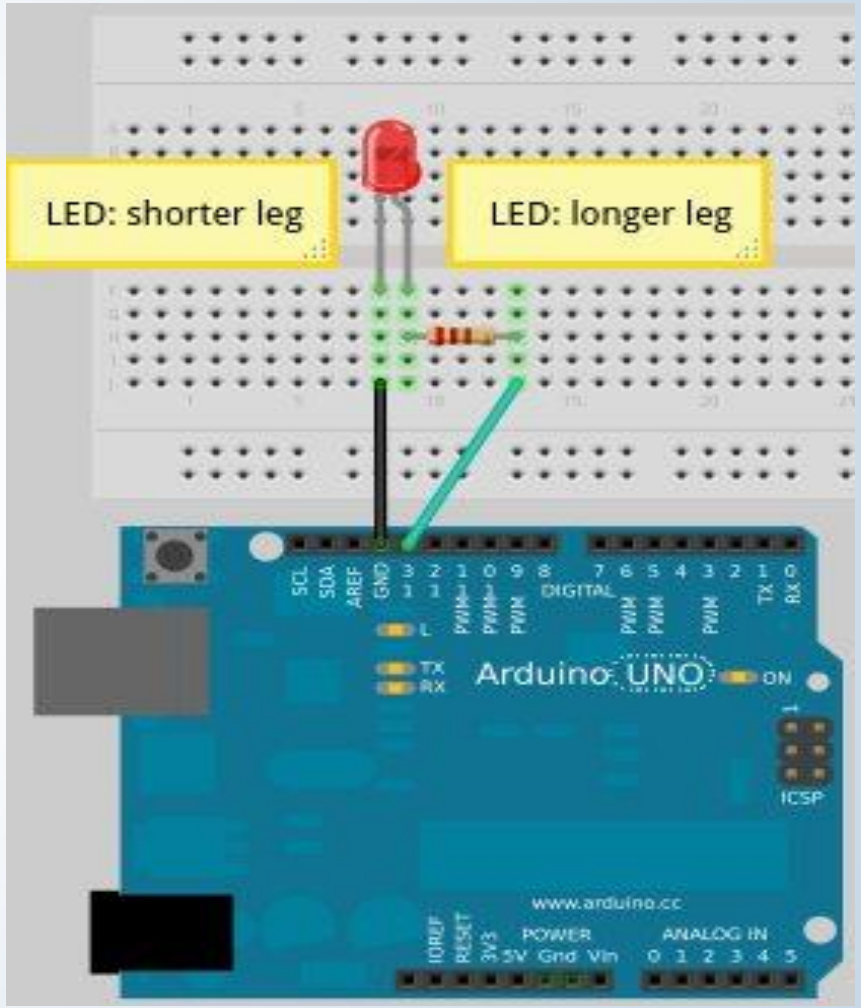
LAB 1: blink LED

```
// LAB 1 - LED Flasher
int ledPin = 13;
void setup()
{
    1. pinMode(ledPin, OUTPUT);
    // OUTPUT means we are sending some data out of  uC
}
void loop()
{
    1. digitalWrite(ledPin, HIGH);
    2. delay(1000);
    3. digitalWrite(ledPin, LOW);
    4. delay(1000);
}
```



LAB 1: blink LED

```
int led1Pin = 13;
int led2Pin = 12;
void setup()
{
    1. pinMode(led1Pin, OUTPUT);
    2. pinMode(led21Pin, OUTPUT);
}
void loop()
{
    1. digitalWrite(led1Pin, HIGH);
    2. digitalWrite(led2Pin, LOW);
    3. delay(1000);
    4. digitalWrite(led1Pin, LOW);
    5. digitalWrite(led2Pin, HIGH);
    6. delay(1000);
}
```

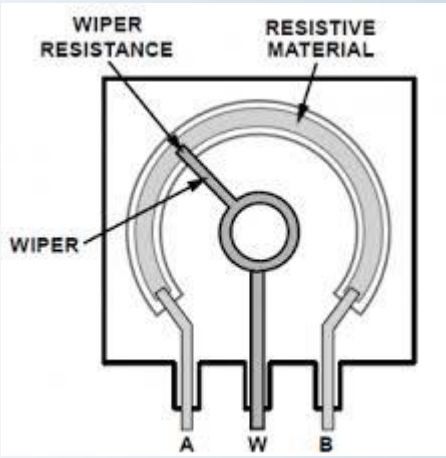


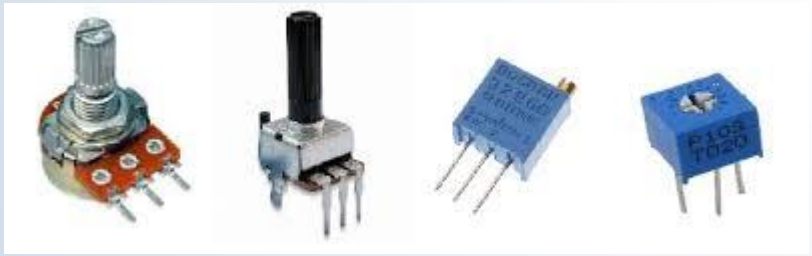
LAB 2: reading the value of analog input => change the blink rate of the LED

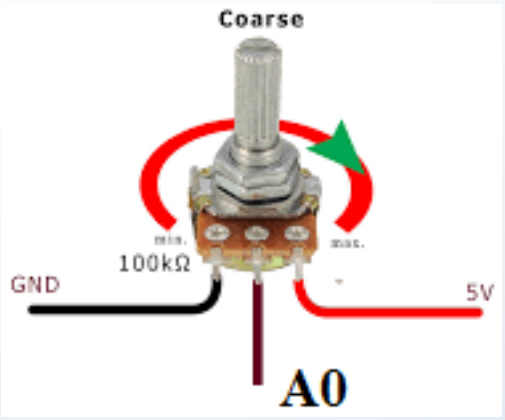
Hardware Required

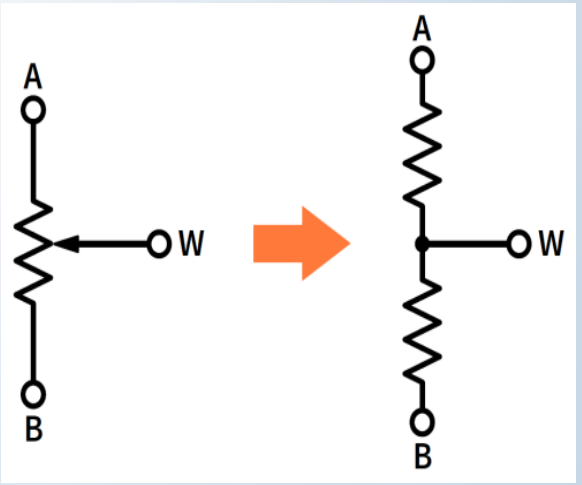
1. variable resistor 10K
2. Led 3mm
3. Wire
4. Breadboard
5. Arduino





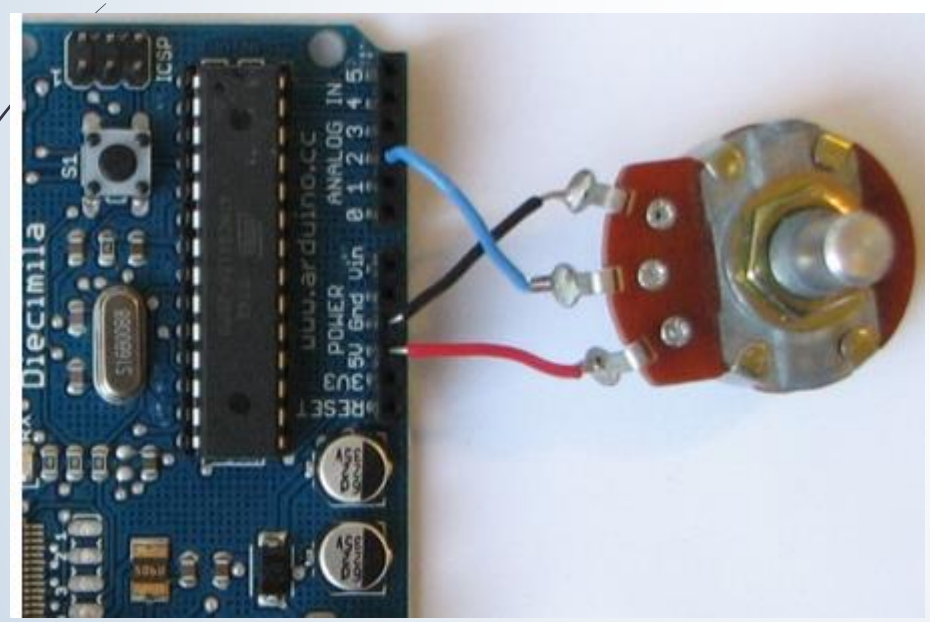






LAB 2: reading the value of analog input => change the blink rate of the LED

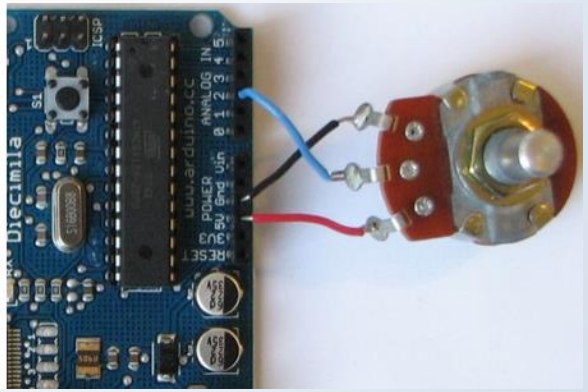
- A potentiometer is a variable resistance, which we can read into the Arduino board as an analog value.




```
// select the input pin for the potentiometer
int potPin = 2;
// variable to store the value coming from the sensor
int potValude = 0;
// read the value from the sensor
potValude = analogRead(potPin);
```

LAB 2: reading the value of analog input and change the blink rate of the LED

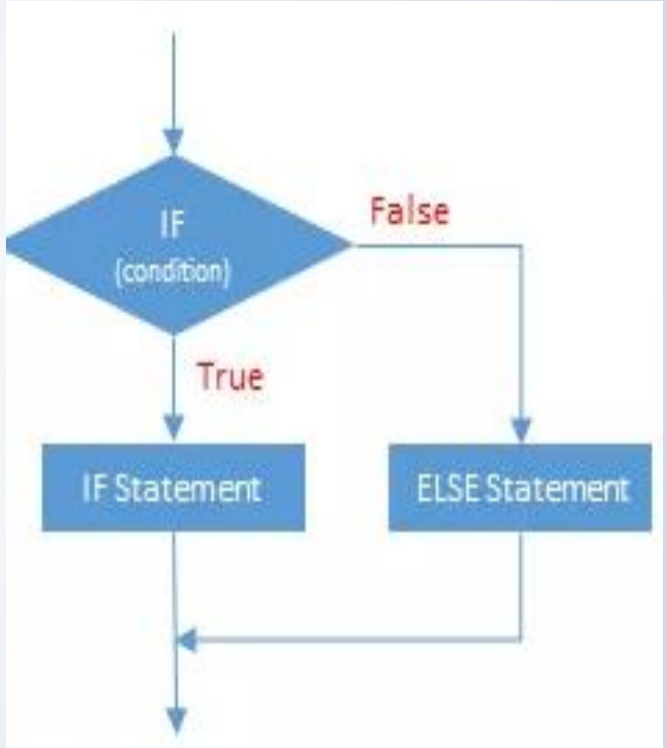
```
// reading the valude of potentiometer
int ledPin = 13;
int potPin = 2;
int potValude = 0;
void setup() {
    1. pinMode(ledPin, OUTPUT);
    2. // pinMode(potPin, INPUT); default Analog pins are INPUT => skip this instruction
}
void loop() {
    1. potValude =  analogRead (potPin);
    2. digitalWrite(ledPin, HIGH);
    3. delay(potValude);
    4. digitalWrite(ledPin, LOW);
    5. delay(potValude);
}
```



LAB 3: If Statement (Conditional Statement)

It allows you to make something happen or not, depending on whether a given condition is true or not.

```
if (someCondition)
 {
     // do stuff if the condition is true
 }
```



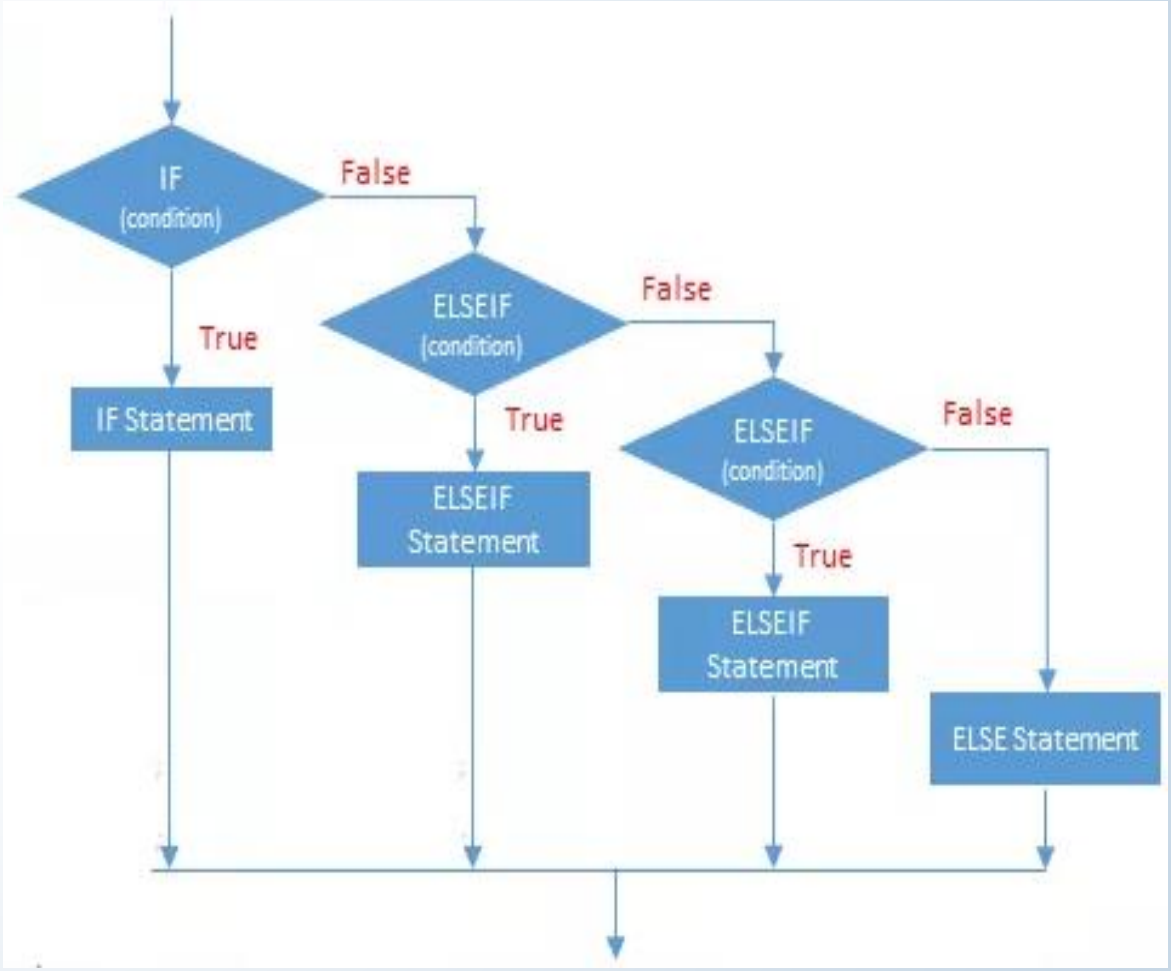
There is a common variation called if-else that looks like this:

```
if (someCondition)
   {
      // do stuff if the condition is true
   }
else
  {
      // do stuff if the condition is false
  }
```

LAB 3: If Statement (Conditional Statement)

There's also the else-if, where you can check a second condition if the first is false:

```
if (someCondition)
{
    // do stuff if the condition is true
}
else if (anotherCondition)
{
   // do stuff only if the first condition is false
   // and the second condition is true
}
```



LAB 3: If Statement (Conditional Statement)

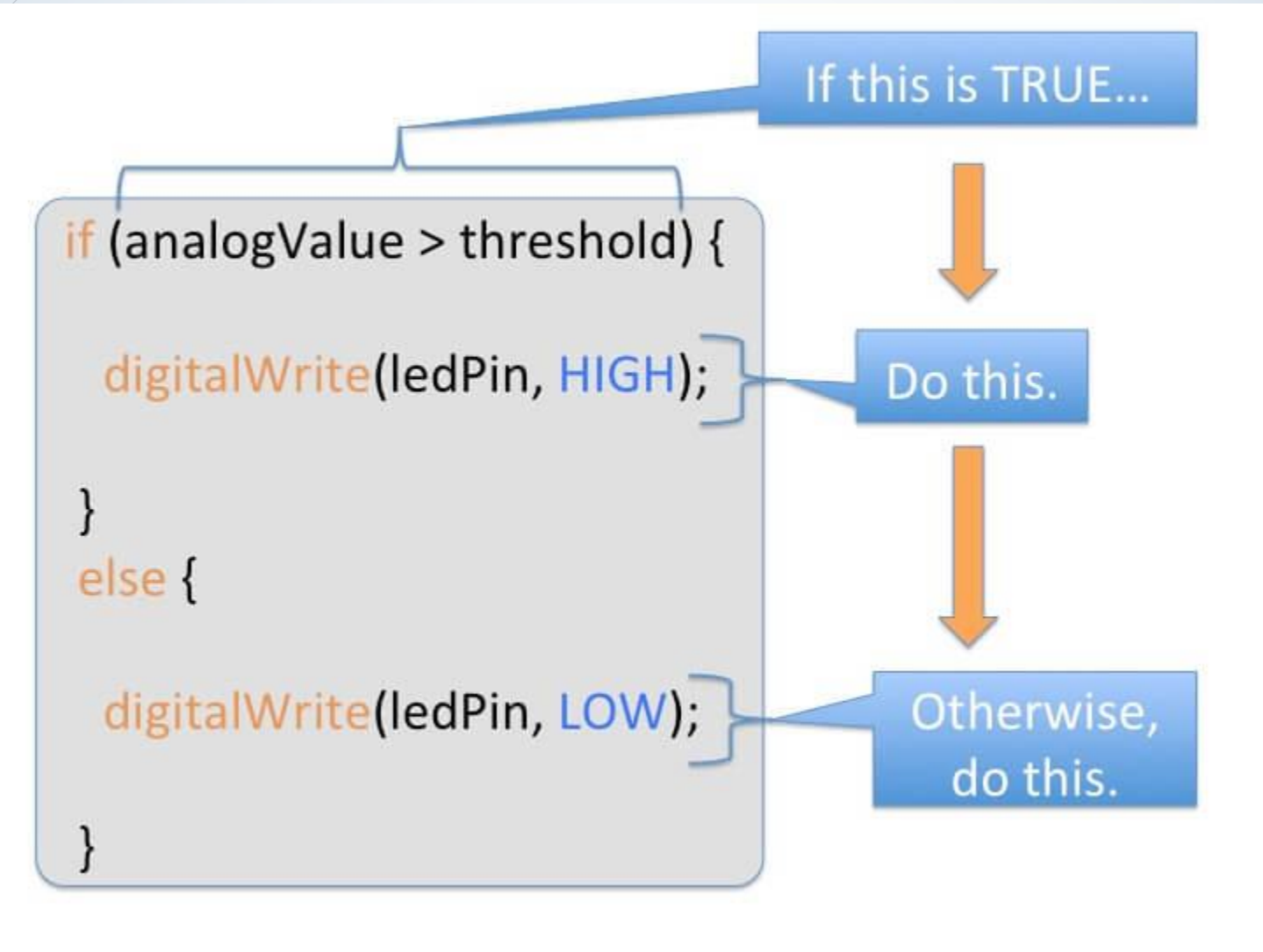
- turns on an LED if the value read on an analog input goes above a certain threshold.

- Hardware Required

1. Arduino or Genuino Board
2. Potentiometer
3. Led 3mm
4. Wire
5. Resistor 330R

LAB 3: If Statement (Conditional Statement)

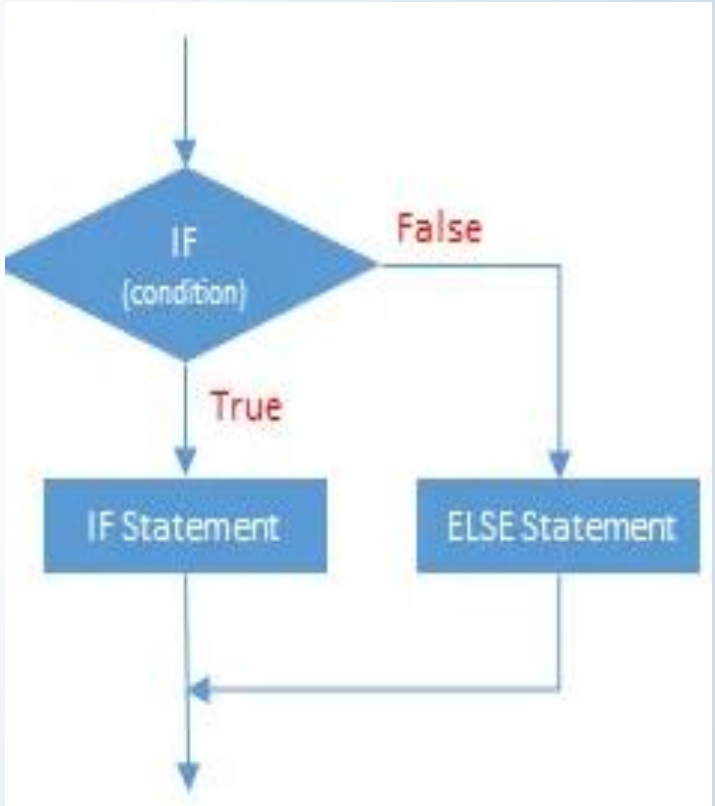
- turns on an LED if the value read on an analog input goes above a certain threshold.



LAB 3: If Statement (Conditional Statement)

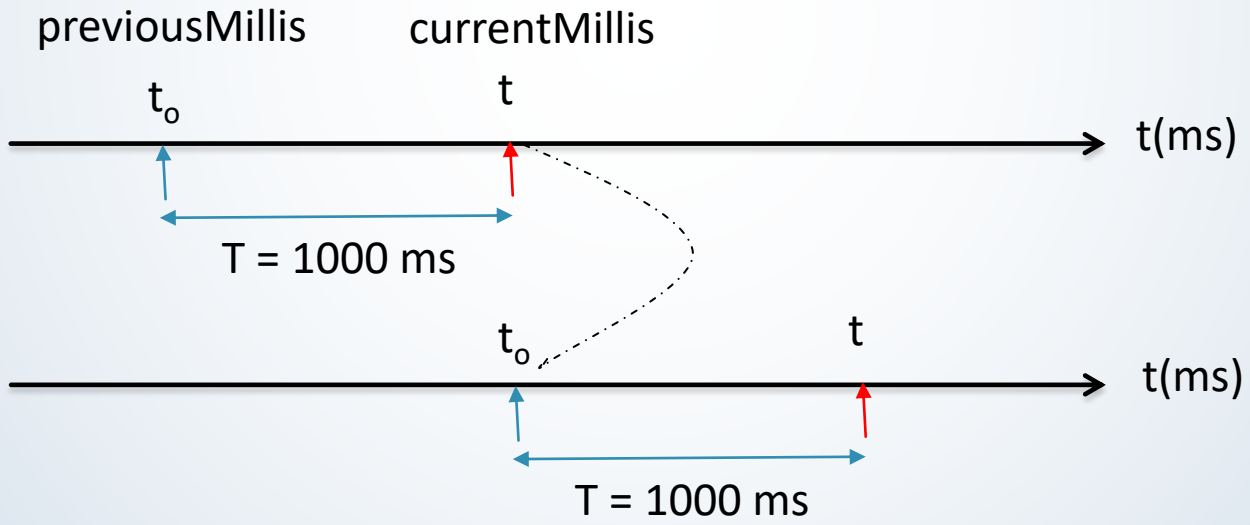
- turns on an LED if the value read on an analog input goes above a certain threshold.

```
const int potPin = A0;
const int ledPin = 13;
const int threshold = 400;  // in the range of the analog input
 void setup()
{
   pinMode(ledPin, OUTPUT);
   Serial.begin(9600);
}
 void loop()
{
   int analogValue = analogRead(potPin);
   if (analogValue > threshold)
      {
         digitalWrite(ledPin, HIGH);
       }
   else
      {
          digitalWrite(ledPin, LOW);
       }
  Serial.println(analogValue); // skip this comment
  delay(1);       // delay in between reads for stability
}
```



LAB 4: Blink Without Delay

- Sometimes you need to do two or many things at once. In this case, you can't use delay(), because Arduino pauses your program during the delay().
- For example: If the button is pressed while Arduino is paused waiting for the delay() to pass, your program will miss the button press.



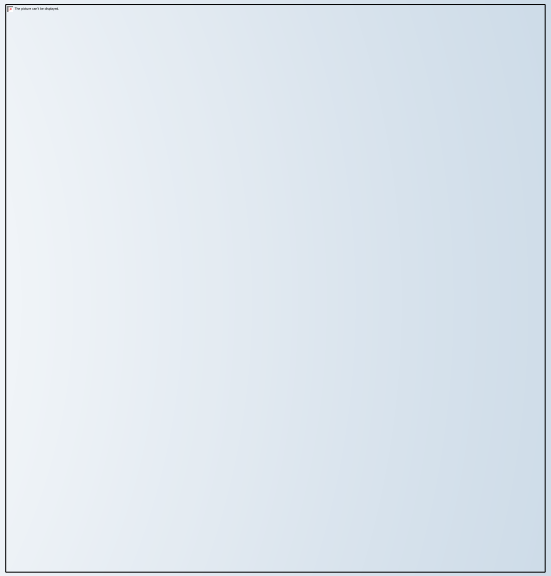
LAB 4: Blink Without Delay

- Hardware Required

1. Arduino UNO
2. LED 3mm
3. 330 ohm resistor







LAB 4: Blink Without Delay

```
1.  const int ledPin =  13;
2.  int ledState = LOW ;
3.  unsigned long previousMillis = 0;
4.  // will store last time LED was updated
5.  const long interval = 1000;
6.  // interval at which to blink (milliseconds)
7.  void setup()
8.  {
9.    pinMode(ledPin, OUTPUT);
10. }
11. void loop()
12. {
13. unsigned long currentMillis = millis();
14. if (currentMillis - previousMillis >= interval)
15.    {
16.      previousMillis = currentMillis;
17.         if (ledState == LOW)
18.    {
19.              ledState = HIGH;
20.           }
21.        else
22.    {
23.              ledState = LOW;
24.           }
25.      digitalWrite(ledPin, ledState);
26.   }
27. }
```

Type Conversion Functions

- • char (), byte ()
- • int (), word (), long ()
- • float (), double ()

CLICK HERE TO KNOW MORE ABOUT DIGITAL & ANALOG INPUTS & OUTPUTS

Digital Input and Output Functions

- • digitalWrite () (writes to a digital output contact)
- • digitalRead () (reads from a digital input contact)

Analog Input and Output Functions

- • analogReference () defines the analogue reference voltage)
- • analogWrite () (writes PWM signals to an output contact)
- • analogRead () (reads by an analog input contact)

Advanced Input and Output Functions

- • tone () (produces a square frequency signal)
- • noTone () (stops producing square signals)
- • shiftOut () (slides the digits of a value into an output contact)
- • pulseIn () (returns the duration to µs of a HIGH or LOW pulse)

Time Functions

- • millis () (run time in ms)
- • micros () (duration of the program in ms)
- • delay () pause - duration is given in ms)
- • delayMicroseconds () (program pause - duration is given in µs)

CLICK HERE TO EXPLORE MORE ABOUT TIME FUNCTIONS

Mathematical and Trigonometric Functions

- • max () (finds the largest of two numbers)
- • min () (finds the smallest of two numbers)
- • abs () (returns the absolute value of a number)
- • constrain () (checks for overflow or margin overflow)
- • map () (performs linear boundary transformation)
- • pow () (returns the result of a force)
- • sqrt () (returns the root of a number)
- • sin () (calculates the sine of a number)
- • cos () (calculates the cosine of a number)
- • tan () (calculates the tangent of a number)

CLICK HERE TO EXPLORE MORE ABOUT MAP()

Pseudo Random Number Generator Functions

Input and Output Functions

- • random () (a new number is given by the generator)
- • randomSeed () (sets the seed of the production generator)

Binary Number Processing Functions

- • lowByte () (returns the right one byte of a variable)
- • highByte () returns the leftmost byte of a variable)
- • bitRead () (reads a specific digit of a variable)
- • bitWrite () (writes to a specific digit of a variable)
- • bitSet () (writes the value 1 to a digit of a variable)
- • bitClear () (writes the value 0 to a digit of a variable)
- • bit () (calculates a specific force based on 2)

Shutdown Service Routines

- • attachInterrupt () (activates a shutdown service routine)
- • detachInterrupt () (disables a shutdown service routine)

Stop Activation and Deactivation Functions

- • interrupts () activates the stop signals
- • noInterrupts () (disables interrupt signals)

Serial Communication Support

- • Serial (object of serial communication with useful methods)

CLICK HERE TO EXPLORE MORE ABOUT SERIAL COMMUNICATION SUPPORT

Type of variables

- boolean, with values 0 and 1 (or True/False)
- byte, with values from 0 to 255
- int, integer with values from -32768 to 32767
- long, integer with values from -2147483648 to 2147483647
- float, decimal numbers
- char, one character (one Byte size)
- string, table of characters

Programming Comments

- As in all programming languages, you can have comments to make it easier to understand and maintain the code you writing. you can use the two vertices // for a comment on a line (the comment follows and is ignored), or you can use the / * .. * / for comments written on multiple lines (that is between / * and * / is ignored) .
- For example:

```
Int ledPin = 13; // set pin number for LED

/ * In the code below we will program one
LED blinks every 1 sec * /
```

Input/Output Port Management Functions (Pins)

- The pinMode (Pin, Mode) function uses its name and arguments **a)** the pin number and **b)** the mode of operation marked by the word INPUT or OUTPUT.
- As is already mentioned we have 14 digital Pins, 6 of which are PWM, with names 0..13 and six analogs with the names A0..A5.
- For example:

```
PinMode(12, OUTPUT);
PinMode(ledPin, OUTPUT);
PinMode(A2, INPUT);
```